Santhosh Reddy Papireddy

Lead Data Scientist | Generative AI, LLMs, RAG & Agentic AI

Bengaluru, India | +91-9505844082 | santhosh49reddy@gmail.com | linkedin.com/in/santhoshreddy39

PROFESSIONAL SUMMARY

Lead Data Scientist with **9+ years** in AI, data science and advanced analytics, including **3+ years leading AI initiatives** and a **team of 7** data scientists and AI engineers. Hands-on in building **GenAI / LLM-powered enterprise solutions** from POC to production: **RAG workflows**, **prompt frameworks**, **AI agents** and **multi-agent orchestration** (LangGraph, AutoGen, Google ADK), with **AI evaluation**, **guardrails** and **Responsible AI** practices. Strong **Python and SQL** on **GCP (Vertex AI, BigQuery)** and **Azure (Azure OpenAI, Azure AI Foundry)**. Translates ambiguous business problems into structured AI workflows, evaluation criteria and implementation-ready specifications, with explainable outputs designed for **human-in-the-loop** review. IIT Madras graduate.

PROFESSIONAL EXPERIENCE

Lead Data Scientist, Accenture (Google XWF) — Bengaluru, India **2024 – Present**

- **Lead and mentor a team of 7** data scientists and AI engineers delivering GenAI, LLM and agentic AI solutions; provide technical direction on architecture, AI quality standards and release readiness, partnering with product owners, architects and business SMEs.

Banking – Agentic AI Deal-Generation Platform

- Architected a **multi-agent Agentic AI** solution using **Google ADK** and **Gemini** on **Vertex AI**, defining the **orchestration layer**, agent inputs/outputs, hand-offs and error scenarios to automate portfolio deal generation for Relationship Manager workflows.
- Designed **structured output schemas** with dynamic schema validation, workflow orchestration and parallel parameter simulation, integrating concurrent banking-system **APIs** to produce auditable recommendations for RM (**human-in-the-loop**) review.
- Cut deal turnaround time **93% (15 days to 1 day)**, maximizing RM throughput and closure speed.

Aviation – GenAI Conversational Support Chatbot

- Led end-to-end delivery of a **Dialogflow CX** chatbot providing 24/7 automated support for a major airline; optimized intent recognition and response orchestration, improving query-resolution accuracy by **25%** at **sub-2-second latency**.

Manufacturing – Enterprise GenAI Chatbot

- Built an enterprise conversational AI chatbot (**Dialogflow CX**) for Samsung Semiconductor's website with multi-turn dialogue flows across products, technologies, support and careers, achieving **95%** query-resolution accuracy.

AI Consultant, KPMG Global Services — Bengaluru, India **2021 – 2024**

Finance – LLM-Driven Query Routing & RAG Solution

- Designed an **LLM-driven GenAI** solution using **RAG architectures**, **prompt engineering** and **model evaluation frameworks** to ground responses in enterprise knowledge and streamline customer inquiries.
- Built LLM-based **query classification** and **entity extraction** producing structured outputs that dynamically populate UI elements by user intent; achieved **100%** categorization and UI-population accuracy on the test set.

Retail – Location Intelligence for Store Expansion

- **Led** a cross-functional location-intelligence solution for UK retail clients, designing **probabilistic models** for store catchment areas and supply-demand and demographic analysis to guide data-driven site selection for business stakeholders.

Associate Analyst, Lennox India Technology Centre — Chennai, India **2017 – 2020**

- Designed end-to-end **ML workflows** across 500+ sensor features, improving fault-detection accuracy to **89%**.
- Built predictive supply-chain forecasting models using **XGBoost** and feature engineering, reducing sales-forecast RMSE to **11.5**.

CORE TECHNICAL SKILLS

Generative AI & LLMs: LLM Strategy & Model Selection, Prompt Engineering, Reusable Prompt Templates, RAG Architectures (Chunking, Retrieval, Grounding), Structured Outputs, GPT, Gemini, Claude, Azure OpenAI, LLM Gateways, Cost-Performance Tuning
Agentic AI & Orchestration: AI Agents, Multi-Agent Systems, Tool Calling, LangGraph, LangChain, AutoGen, CrewAI, LlamaIndex, Google ADK
AI Evaluation & Governance: LLM Evaluation Frameworks, Model Evaluation, Guardrails, Hallucination Mitigation, Human-in-the-Loop, Explainability, Responsible AI, AI Development Lifecycle (AIDLC)
Cloud & Data Platforms: GCP (Vertex AI, Vertex AI Pipelines, BigQuery, Cloud Storage, Dialogflow CX), Azure (Azure OpenAI, Azure AI Foundry, Azure ML Studio), Vector Databases (FAISS, Pinecone, Chroma)
Machine Learning: Regression, Classification, Clustering, XGBoost, Deep Learning, Transformers, TensorFlow, PyTorch, Scikit-learn, NLP, Time Series Forecasting, Geospatial Analysis, Statistics
Programming & Tools: Python, SQL, FastAPI, Flask, REST APIs, Streamlit, Git, MLOps

EDUCATION

M.Tech, Applied Mechanics — IIT Madras, Chennai, India **2015 – 2017**

B.Tech, Mechanical Engineering — JNTU Anantapur, India **2009 – 2013**

CERTIFICATIONS & AWARDS

Certifications: Google Generative AI Leader; Google Associate Data Practitioner; Microsoft Build/Operate ML Solution
Awards: Multiple Quarterly and Bi-Monthly Awards for excellence in AI solution delivery, ethical AI practices and project execution